

MUHAMMAD IDREES

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PROFESSIONAL SUMMARY

AI Engineer with hands-on, project-based experience across Machine Learning, Deep Learning, Computer Vision, NLP and Generative AI/Agentic AI. Skilled at breaking down complex models and pipelines into clear, practical explanations, and experienced presenting technical evaluations to stakeholders. Built and deployed real-world AI systems spanning document intelligence, computer vision, cybersecurity analytics and predictive modeling practical, industry-grounded work suited to hands-on labs and project-based training. Bachelor's in Artificial Intelligence, with a published research paper and multiple AI/ML certifications.

CORE AI/ML & TRAINING-RELEVANT SKILLS

- **Programming & Data:** Python, Pandas, NumPy, EDA, Data Preprocessing, Feature Engineering, Data Validation
- **Machine Learning:** Scikit-learn, Random Forest, XGBoost, KNN, Logistic Regression, Decision Trees, Isolation Forest
- **Deep Learning:** TensorFlow, Keras, CNN, ResNet, LSTM, BiLSTM, Attention Mechanisms, Transformers, Autoencoders
- **Computer Vision & OCR:** OpenCV, YOLO, Object Detection, Image Classification, Pose Estimation, DeepSORT, OCR-aware document parsing, Roboflow
- **Generative AI & Agentic AI:** LLMs, RAG, Prompt Engineering, AI Agents, LangChain, LangGraph, n8n, Tool Calling, Vector Databases (ChromaDB)
- **AI Engineering & Deployment:** FastAPI, REST APIs, PostgreSQL, AI Pipelines, Workflow Automation, Streamlit, Docker, Git/GitHub, Jupyter, Google Colab
- **Communication & Knowledge-Sharing:** Technical presentation of model evaluations to stakeholders, project-based problem solving, research writing and publication

PROFESSIONAL EXPERIENCE

AI Engineer — National Center of Artificial Intelligence (NCAI), UET Peshawar Feb-2025 -- Aug-2025

AI Engineer — ExatecSolution Peshawar Aug-2027 -- Present

- Present technical model evaluations to project stakeholders, translating results into clear, actionable insights — direct experience explaining AI concepts to a non-implementing audience.
- Design and build end-to-end AI/ML pipelines data cleaning, EDA, feature engineering and validation the practical workflow trainees need to see applied from raw data to working model.
- Performed hyperparameter optimization and structured model experimentation to improve predictive performance and efficiency.

AML PROJECTS

- AI Knowledge Assistant — Advanced RAG Pipeline for Document Q&A: Full-stack RAG system with hybrid retrieval, reranking, OCR-aware parsing, and LLM-based source-cited answers a strong basis for teaching RAG and document-intelligence concepts.
- Bone Fracture Detection System: YOLOv8 object detection for X-ray fracture identification, achieving 99.5% mAP@50 demonstrates applied computer vision suitable for CV labs.
- Network Traffic Anomaly Detection System: XGBoost, Random Forest, LSTM, Isolation Forest, and Autoencoder models, reaching 99.59% accuracy and 99.96% ROC-AUC with XGBoost useful for teaching applied ML model comparison.

RESEARCH & PUBLICATIONS

- Published Research Paper: "Enhancing K-Nearest Neighbors for Categorical Data Classification Using Similarity-Based Distance Metrics" reflects the ability to research, structure, and communicate technical AI concepts in writing.

EDUCATION

Bachelor of Science in Artificial Intelligence

University of Agriculture, Peshawar | CGPA: 3.40 / 4.00

CERTIFICATIONS

Agentic AI (Edureka)
The Art of Prompt Engineering (IBM)
Prompt Engineering (Vanderbilt University)
Machine Learning, AI & Data Science (E-Hunar)
Feature Engineering for Machine Learning (Udemy)

ACHIEVEMENTS

Languages: English (Professional Working Proficiency) | Urdu (Native) | Pashto (Native)

REFERENCES

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